

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405647
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sun; Yongfu et al.

Mobile terminal and middle frame assembly

Abstract

A mobile terminal includes middle frame assembly that is configured to bear an electronic component in the mobile terminal. The electronic component includes a heat source, and the middle frame assembly includes a middle frame, one or more heat pipes, and a first vapor chamber. The middle frame includes a heat dissipation region corresponding to the heat source. The first vapor chamber is accommodated in the heat dissipation region. The heat pipes are connected to the first vapor chamber, and are configured to dissipate heat from the first vapor chamber. This structure of the mobile terminal and the middle frame assembly of the mobile terminal improves heat dissipation performance of the mobile terminal.

Inventors: Sun; Yongfu (Shenzhen, CN), Yang; Jie (Dongguan, CN), Shi; Jian (Shanghai, CN)

Applicant: HUAWEI TECHNOLOGIES CO., LTD. (Guangdong, CN)

Family ID: 77891811

Assignee: HUAWEI TECHNOLOGIES CO., LTD. (Shenzhen, CN)

Appl. No.: 17/951274

Filed: September 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230022994 A1	Jan. 26, 2023

Foreign Application Priority Data

CN	202010217496.3	Mar. 24, 2020
CN	202020393505.X	Mar. 24, 2020

Related U.S. Application Data

continuation parent-doc WO PCT/CN2021/082689 20210324 PENDING child-doc US 17951274

Publication Classification

Int. Cl.: G06F1/20 (20060101); H01Q1/02 (20060101); H05K7/20 (20060101)

U.S. Cl.:

CPC G06F1/203 (20130101); H01Q1/02 (20130101); H05K7/20336 (20130101);

Field of Classification Search

CPC: H05K (7/2039); H05K (7/20518); H05K (7/20309); H05K (7/2033); G06F (1/203); H01Q (1/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9013874	12/2014	Kaldani	361/679.52	F28F 3/048
9625215	12/2016	Hsiao	N/A	G06F 1/203
10605538	12/2019	Ahamed	N/A	F28D 15/046
2012/0132402	12/2011	Aoki	165/104.21	F28D 15/06
2013/0198501	12/2012	Wu	713/2	H01L 23/427
2014/0321058	12/2013	Fujieda	29/890.032	F28D 15/0233
2015/0253823	12/2014	Han	N/A	N/A
2017/0235349	12/2016	Ghioni et al.	N/A	N/A
2017/0292793	12/2016	Sun	N/A	F28F 9/001
2018/0142961	12/2017	Wu	N/A	F28D 15/0266
2018/0373299	12/2017	Wang et al.	N/A	N/A
2020/0323103	12/2019	Lee	N/A	F28D 15/046
2022/0087067	12/2021	Jia	N/A	H05K 5/0086

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104039115	12/2015	CN	N/A
106304817	12/2016	CN	N/A
106993393	12/2016	CN	N/A
206441100	12/2016	CN	N/A
104488371	12/2016	CN	N/A
108513515	12/2017	CN	H05K 7/20336
108601293	12/2017	CN	N/A

108617159	12/2017	CN	N/A
108650866	12/2017	CN	N/A
209358916	12/2018	CN	N/A
209563090	12/2018	CN	N/A
110794641	12/2019	CN	N/A
210183768	12/2019	CN	N/A
212164015	12/2019	CN	N/A
102077002	12/2019	KR	N/A
2019103327	12/2018	WO	N/A

OTHER PUBLICATIONS

CN-108513515-A Translation (Year: 2018). cited by examiner
Extended European Search Report dated Jul. 11, 2023 issued for European Application No. 21774280.8 (11 pages). cited by applicant
International Search Report dated May 11, 2021 issued for International Application No. PCT/CN2021/082689 (10 pages). cited by applicant

Primary Examiner: Gandhi; Jayprakash N

Assistant Examiner: Muir; Matthew Sinclair

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/082689, filed on Mar. 24, 2021, which claims priority to Chinese Patent Application No. 202010217496.3, filed on Mar. 24, 2020 and Chinese Patent Application No. 202020393505.X, filed on Mar. 24, 2020. The disclosures of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

(1) This disclosure relates to the field of terminal technologies, and in particular, to a mobile terminal and a middle frame assembly.

BACKGROUND

(2) Heat is separately dissipated from all existing mobile terminals by using a single heat plate or a single heat pipe. However, as the mobile terminal becomes intelligent at an increasingly high degree, more heat is generated in a main frequency upgrade of the mobile terminal. If excessive heat cannot be rapidly dissipated, not only a service life of the mobile terminal is affected, but also intuitive experience of a consumer is affected.

SUMMARY

(3) Embodiments of this disclosure provide a terminal and a middle frame assembly thereof, to improve heat dissipation performance of a mobile terminal.

(4) According to a first aspect, the present invention provides a middle frame assembly, configured to bear an electronic component in a mobile terminal. The electronic component includes a heat source, and the middle frame assembly includes a middle frame, one or more heat pipes, and a first vapor chamber. The middle frame includes a heat dissipation region corresponding to the heat source, and the first vapor chamber is accommodated in the heat dissipation region. The first vapor chamber may be further fixedly connected to the middle frame by using the heat dissipation region, and a specific fastening manner is not limited to the following several manners: The first vapor chamber may be fastened to the middle frame through bottom welding, may be fastened to the

middle frame through side welding, may be fastened to the middle frame through bottom bonding, may be fixedly connected to the middle frame by using circumferential overlapping structures, or may be connected to the middle frame through combination of bottom surface fastening and circumferential fastening. Each heat pipe is connected to the first vapor chamber (a manner of connecting the heat pipe and the first vapor chamber is not limited to fastening through overlapping and welding, sealing through welding after the heat pipe and the first vapor chamber are connected, or the like), and is configured to dissipate heat from the first vapor chamber. In this disclosure, one end of the heat pipe is connected to the first vapor chamber, and heat of the heat source may be rapidly dissipated by using a comprehensive heat conduction effect of two-dimensional extended heat conduction of the first vapor chamber and one-dimensional linear heat conduction of the heat pipe, to prolong a service life of the heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer.

(5) In a possible implementation, at least one of the one or more heat pipes is further connected to a part that is in the middle frame and whose temperature is lower than a temperature of the first vapor chamber. The part that is in the middle frame and whose temperature is lower than the temperature of the first vapor chamber may be a region at an edge of the middle frame, such as edges on two sides of the middle frame or an edge at the bottom of the middle frame. In this way, when the mobile terminal runs normally, because the first vapor chamber is fastened to the middle frame, heat may be transferred to the first vapor chamber and transferred to the heat pipe, the heat transferred to the heat pipe may be transferred to the entire middle frame, heat of the heat source may be conducted to the entire middle frame, and the heat is conducted out by using the middle frame. An area of the middle frame is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal.

(6) In a possible implementation, at least one of the one or more heat pipes is connected to the middle frame by using a heat conducting medium. The heat conducting medium may be a heat conducting adhesive or a graphite material. Heat conductivity of the heat conducting adhesive is twice over heat conductivity of an ordinary adhesive layer. The heat conducting adhesive has good heat conduction performance, and can rapidly transfer heat on the heat pipe to the entire middle frame, so that heat dissipation efficiency is high.

(7) In a possible implementation, internal space of at least one of the one or more heat pipes is connected to internal space of the first vapor chamber to form a heat conducting cavity. A relatively large and complete heat conducting cavity is formed inside after the heat pipe and the first vapor chamber are connected. Working fluid in the heat conducting cavity passes through a longer path and takes longer time to complete one time of heat conduction circulation, so that heat of the heat source may be transferred to a larger region of the middle frame, and the heat is conducted out more dispersedly, and therefore heat dissipation efficiency is high.

(8) In a possible implementation, at least one of the one or more heat pipes is inserted into the first vapor chamber. In this manner, the heat pipe overlaps the first vapor chamber inside the first vapor chamber. Connectivity between the heat pipe and the first vapor chamber is better.

(9) In a possible implementation, the electronic component further includes a battery spaced from the heat source, and the middle frame further includes a battery placement region corresponding to the battery. The battery placement region includes a first edge and a second edge that are disposed opposite to each other, the heat dissipation region is adjacent to the first edge, and at least one of the one or more heat pipes extends from the heat dissipation region to a direction of the second edge. In this way, the heat pipe may extend in a larger range of the middle frame, to transfer heat of the heat source to a further distance, so that a heat dissipation effect is fairly good.

(10) In a possible implementation, at least one of the one or more heat pipes is not in contact with the battery. Therefore, when the mobile terminal runs normally, the mobile terminal generates heat, and a temperature of a part that is in the middle frame and that is connected to the heat pipe is less than that of the first vapor chamber. When the heat pipe is not in contact with the battery, the heat

pipe and the battery do not transfer heat to each other, and the heat pipe and the battery are not interfered with each other. In addition, the heat pipe is disposed avoiding the battery, and space for placing the heat pipe does not need to be disposed on a surface of the battery, which facilitates a thin-type design of the mobile terminal.

(11) In a possible implementation, the one or more heat pipes include a first heat pipe and a second heat pipe that are disposed at intervals, the first heat pipe surrounds the battery placement region, and the second heat pipe is disposed overlapping the battery placement region. In this implementation, the heat pipe is disposed in both the battery placement region and a periphery of the battery placement region, to enhance heat dissipation performance.

(12) In a possible implementation, the middle frame includes a top, a bottom disposed opposite to the top, and two sidewalls located between the top and the bottom. The heat dissipation region is located between the first edge of the battery placement region and the top, and another element placement region is disposed between the second edge of the battery placement region and the bottom. The middle frame further includes a second vapor chamber, the second vapor chamber is fastened in the another element placement region, and at least one of the one or more heat pipes is further connected to the second vapor chamber. When the heat source is fixedly placed in the heat dissipation region and another heat source is fixedly placed in the another element placement region, heat of the heat source may be conducted to the first vapor chamber and conducted to the heat pipe, and heat of the another heat source may be conducted to the second vapor chamber and conducted to the heat pipe. Further, the heat of the heat source and the another heat source may be rapidly dissipated under comprehensive action of the first vapor chamber, the second vapor chamber, and the heat pipe, to prolong a service life of the heat source and that of the another heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, because the first vapor chamber and the second vapor chamber are fastened to the middle frame, the heat of the heat source and the another heat source may be conducted to the entire middle frame, and the heat is conducted out by using the middle frame, thereby implementing heat dissipation performance of the entire middle frame. An area of the middle frame is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal.

(13) In a possible implementation, one end of at least one of the one or more heat pipes overlaps the first vapor chamber, and the other end of the at least one of the one or more heat pipes overlaps the second vapor chamber. This implementation provides a manner of connecting the heat pipe to the first vapor chamber and the second vapor chamber. Overlapping means that in a direction perpendicular to the first vapor chamber and the second vapor chamber, there is a region in which the heat pipe partially overlaps the first and second vapor chambers. This connection structure is stable and has good reliability.

(14) In a possible implementation, at least one of the one or more heat pipes extends along the periphery of the battery placement region, and at least a part of the heat pipe is located in a gap between the battery placement region and a sidewall of the middle frame. The gap between the battery placement region and the sidewall of the middle frame is idle space. In this disclosure, at least a part of the heat pipe is accommodated in the gap between the battery placement region and the sidewall of the middle frame, that is, a part of the heat pipe is accommodated in the idle space. In this way, not only the idle space is used, but also no new space needs to be disposed in the mobile terminal to accommodate the heat pipe, so that a volume of the mobile terminal is not increased.

(15) In a possible implementation, at least one of the one or more heat pipes is in contact with the surface of the battery. Therefore, heat on the battery may be further transferred to the heat pipe and is then transferred to the entire middle frame, so that the middle frame dissipates the heat from the battery, and a heat dissipation area is large, and a heat dissipation effect is fairly good.

(16) In a possible implementation, the one or more heat pipes include at least a first heat pipe and a

second heat pipe that are disposed at intervals, and both the first heat pipe and the second heat pipe are disposed around the battery placement region. In addition, when both the first heat pipe and the second heat pipe are disposed around the battery placement region, space between the battery placement region and the sidewall is used, and neither of the two heat pipes overlaps the battery placement region and affects a thickness of the middle frame. When the battery is placed in the mobile terminal, the heat pipe does not overlap the battery, and does not affect the thickness of the mobile terminal.

(17) In a possible implementation, at least one of the one or more heat pipes includes at least a first heat pipe and a second heat pipe that are disposed at intervals, the first heat pipe surrounds the battery placement region, and the second heat pipe overlaps the battery placement region. In this way, heat dissipation effects of the two heat pipes can be used, a region that is in the middle frame and that can be used for heat dissipation is larger, and a heat dissipation effect is better, so that heat of the heat source can be dissipated more rapidly, to prolong a service life of the heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, space between the battery placement region and the sidewall of the middle frame and other space can be effectively used. In addition, the manner in which the second heat pipe overlaps the battery placement region facilitates installation of the second heat pipe and is convenient for operation, and the second heat pipe may be configured to dissipate heat of the battery.

(18) In a possible implementation, the middle frame is provided with a first removed region and a third removed region, and the third removed region is formed at the bottom of the first removed region. The first vapor chamber is embedded in the first removed region, at least one of the one or more heat pipes is partially embedded in the third removed region, and a part of the at least one of the one or more heat pipes in the third removed region is stacked with the first vapor chamber. The first removed region is a groove or a hole, and the second removed region is a groove or a hole. In this implementation, a part of the heat pipe is located between the first vapor chamber and the heating source. When the first removed region and the second removed region have groove structures, openings of the first removed region and the second removed region face a side that is of the middle frame and that is away from the heating source. The heating source is spaced from the heat pipe and the first vapor chamber by the middle frame, and the middle frame is configured to support the heating source.

(19) In a possible implementation, the middle frame is provided with a first removed region, a second removed region, and a third removed region. The first vapor chamber is embedded in the first removed region, the second vapor chamber is embedded in the second removed region, and at least one of the one or more heat pipes is embedded in the third removed region. The first removed region is a groove or a hole, the second removed region is a groove or a hole, and the third removed region is a groove or a hole. In this manner, the vapor chamber and the heat pipe are embedded in the removed regions, so that the first vapor chamber, the second vapor chamber, and the heat pipe do not increase a thickness of the middle frame, and the mobile terminal may be relatively thin. It may be understood that the first removed region, the second removed region, and the third removed region may be simultaneously disposed in one implementation. In another implementation, one or two of the three removed regions may be selectively configured based on a specific structure in the mobile terminal. For example, in some places, a gap exists between electronic components and can be used for placing the heat pipe. In this case, the third removed region does not need to be disposed. This architecture facilitates a thin-type design of the entire mobile terminal, and can reduce a weight of the mobile terminal.

(20) In a possible implementation, that at least one of the one or more heat pipes is connected to the first vapor chamber means that internal space of the heat pipe and that of the first vapor chamber are interconnected to form a relatively large heat conducting cavity. A relatively large and complete heat conducting cavity is formed inside after the heat pipe and the first vapor chamber are

connected. Working fluid in the heat conducting cavity passes through a longer path and takes longer time to complete one time of heat conduction circulation, so that heat of the heat source may be transferred to a larger region of the middle frame, and the heat is conducted out more dispersedly, and therefore heat dissipation efficiency is high. The heat pipe and the second vapor chamber may also be disposed to form a connection architecture.

(21) In a possible implementation, a thickness of the first vapor chamber is increased step by step, and/or a thickness of the second vapor chamber is increased step by step, to correspondingly adapt to an electronic component whose height is decreased step by step.

(22) In a possible implementation, quantities of heat pipes, first vapor chambers, and second vapor chambers are not limited to one, and there may be two or more heat pipes, first vapor chambers, and second vapor chambers, which may be configured based on a specific heat dissipation requirement in the mobile terminal.

(23) In a possible implementation, a heat conducting film is disposed on an outer surface of any one or more of at least of the one or more heat pipes, the first vapor chamber, and the second vapor chamber. The heat conducting film is disposed, so that heat on the heat pipe and the vapor chamber is conducted to the outside more rapidly.

(24) In a possible implementation, the middle frame is provided with a groove, the first vapor chamber is accommodated in the groove, and the first vapor chamber is fastened to a bottom of the groove by using a heat conducting adhesive. In a specific implementation, the middle frame includes a first overlapping side extending toward the inside of the groove, the first vapor chamber includes a third overlapping side, and the first overlapping side is fastened to the third overlapping side.

(25) It may be understood that the middle frame may further include a second overlapping side configured to fasten the second vapor chamber. The first vapor chamber includes the third overlapping side, and the first overlapping side overlaps the third overlapping side, so that the first vapor chamber is fastened in the heat dissipation region. The first overlapping side and the third overlapping side may be fastened through welding. Similarly, the second vapor chamber may include a fourth overlapping side, and the second overlapping side overlaps the fourth overlapping side, so that the second vapor chamber is fastened in the another element placement region. This fastening manner is not only easy to implement, but also has a good fastening effect.

(26) Specifically, in an implementation, in the solution in which the first vapor chamber is fixedly connected to the middle frame by using an overlapping side, the groove provided on the middle frame may be a hole. When the groove is a hole, the middle frame has no part used to bear the first vapor chamber, but the first vapor chamber may be connected to the middle frame by using an overlapping side.

(27) In another implementation, a first groove provided on the middle frame may be a blind hole, that is, the first groove is not a hole. In this way, a partial structure, which may be referred to as a bearing part, exists in the middle frame, and may be used to bear the first vapor chamber. The first vapor chamber is fastened to the bearing part of the middle frame by using a heat conducting adhesive. In this implementation, to reduce a thickness of the heat conducting adhesive, after the thickness of the heat conducting adhesive becomes smaller, strength of connection between the first vapor chamber and the middle frame becomes insufficient. To protect connection strength and reliability, an overlapping structure may also be disposed, that is, the first overlapping side on the middle frame cooperates with the third overlapping side on the first vapor chamber to enhance reliability of connection between the first vapor chamber and the middle frame. In this implementation, the first vapor chamber and the middle frame are fastened through cooperation of the heat conducting adhesive and the overlapping structure.

(28) In a possible implementation, at least one of the one or more heat pipes extends between electronic components, and/or at least one of the one or more heat pipes extends between the electronic component and a boundary of the middle frame. In this way, not only a gap between the

electronic components and a gap between the electronic component and the boundary of the middle frame can be used to accommodate the heat pipe, but also no new space needs to be added to accommodate the heat pipe, so that a thickness of the mobile terminal does not need to be increased.

(29) This disclosure provides a mobile terminal, including a main board and the foregoing middle frame. The main board includes the electronic component, the main board is installed in the middle frame, and the heat source of the main board is disposed in a heat dissipation region of the middle frame. Heat on the mobile terminal may be rapidly dissipated, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, because the first vapor chamber is fastened to the middle frame, heat of the heat source may be conducted to the entire middle frame, and the heat is conducted out by using the middle frame, thereby implementing heat dissipation performance of the entire middle frame. An area of the middle frame is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal.

(30) In a possible implementation, the mobile terminal further includes a screen, a main board, a battery, and a battery cover, and the middle frame further includes a battery placement region corresponding to the battery. The battery placement region is disposed on one side of the heat dissipation region, the screen and the main board are respectively installed on two opposite sides of the middle frame, the heat source of the main board is disposed in the heat dissipation region of the middle frame, the battery is electrically connected to the main board, the battery is located on a side that is of the main board and that is away from the middle frame, and the battery is located in the battery placement region. The battery cover is installed on the middle frame, and the battery cover fastens the main board and the battery in the middle frame.

(31) In a possible implementation, a wire groove is disposed on an outer surface of the heat pipe, the wire groove is configured to fasten an internal conducting wire of the mobile terminal, and the conducting wire is configured to electrically connect electronic components in the mobile terminal. For example, the conducting wire is configured to electrically connect an electronic component in the heat dissipation region and an electronic component in the another element placement region. The wire groove on the surface of the heat pipe may be used as a carrier of the conducting wire, so that the conducting wire may be fastened in the wire groove of the heat pipe, to fasten a location of the conducting wire, thereby avoiding damage caused to the conducting wire due to shaking, and prolonging a service life of the mobile terminal.

(32) According to a second aspect, this disclosure further provides a mobile terminal, including a main board and the middle frame assembly according to any of the possible implementations. The heat source is disposed on the main board, and the main board is installed in the middle frame.

(33) According to a third aspect, this disclosure further provides a heat dissipation assembly, including at least one vapor chamber and at least one heat pipe in the first aspect and the implementations of the first aspect. For a manner of connecting the at least one vapor chamber and the at least one heat pipe, refer to the foregoing aspect and the implementation. Details are not described herein again. As described in the foregoing aspect, the heat dissipation assembly provided in this aspect may cooperate with a middle frame in a mobile terminal to implement better heat dissipation. In another implementation, the heat dissipation assembly may cooperate with another component in another device to implement better heat dissipation.

(34) In conclusion, in this disclosure, one end of the heat pipe is connected to the first vapor chamber, and heat of the heat source may be rapidly dissipated by using a comprehensive heat conduction effect of two-dimensional extended heat conduction of the vapor chamber and one-dimensional linear heat conduction of the heat pipe, to prolong a service life of the heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, because the first vapor chamber is fastened to the middle frame, heat of the heat source may be conducted to the entire middle frame, and the heat is conducted out by using

the middle frame, thereby implementing heat dissipation performance of the entire middle frame. An area of the middle frame is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) To describe the technical solutions in embodiments of this disclosure or in the background more clearly, the following briefly describes the accompanying drawings for describing embodiments of this disclosure or the background.
- (2) FIG. 1a is a schematic cross-sectional diagram of a mobile terminal according to an embodiment of the present invention;
- (3) FIG. 1b is another schematic cross-sectional diagram of a mobile terminal according to an embodiment of the present invention;
- (4) FIG. 2 is a schematic top view of a middle frame in a middle frame assembly according to an embodiment of the present invention;
- (5) FIG. 3 is a schematic top view of a first structure of a middle frame assembly according to an embodiment of the present invention;
- (6) FIG. 4 is a schematic top view of a second structure of a middle frame assembly according to an embodiment of the present invention;
- (7) FIG. 5 is a schematic cross-sectional diagram in which a heat pipe overlaps a first vapor chamber;
- (8) FIG. 6 is a schematic cross-sectional diagram in which a heat pipe is connected to a first vapor chamber;
- (9) FIG. 7 is a schematic cross-sectional diagram in which a thickness of a first vapor chamber is increased step by step;
- (10) FIG. 8 is a schematic top view of a third structure of a middle frame according to an embodiment of the present invention;
- (11) FIG. 9 is a schematic cross-sectional diagram in which a heat pipe overlaps a first vapor chamber and a second vapor chamber;
- (12) FIG. 10 is a schematic cross-sectional diagram in which a heat pipe is connected to a first vapor chamber and a second vapor chamber;
- (13) FIG. 11 is a schematic top view of a fourth structure of a middle frame assembly according to an embodiment of the present invention;
- (14) FIG. 12 is a schematic top view of a fifth structure of a middle frame assembly according to an embodiment of the present invention; and
- (15) FIG. 13 is a schematic top view of a sixth structure of a middle frame assembly according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

- (16) The following describes embodiments of this disclosure with reference to the accompanying drawings in embodiments of this disclosure.
- (17) Refer to FIG. 1a, FIG. 1b, and FIG. 2. FIG. 1a and FIG. 1b are two schematic cross-sectional diagrams of a mobile terminal according to an embodiment of the present invention. The mobile terminal includes but is not limited to devices such as a tablet computer, a mobile phone, a watch, an electronic reader, a remote control, a personal computer (PC), a notebook computer, an in-vehicle device, a network television, and a wearable device. The mobile terminal may include a main board 10, a screen 20, a battery 30, a middle frame assembly 40, and a battery cover 110.
- (18) The main board 10 may include a plurality of electronic components. Some electronic components (for example, various processors) generate a relatively large amount of heat during

working, and the heat needs to be dissipated from the electronic components (for example, some heat dissipation components are installed). These electronic components are referred to as heat sources in this disclosure.

(19) As shown in FIG. 1b, the middle frame assembly **40** may include a middle frame **401**, a heat pipe **60**, and a first vapor chamber (VC) **70**. The middle frame **401** is an internal bracket of a device and may be configured to bear the screen **20**, the main board **10**, and another electronic component. The screen **20** and the main board **10** are respectively installed on two opposite sides of the middle frame **401**. The battery **30** and the main board **10** are located on a same side of the middle frame **401**. The battery **30** is electrically connected to the main board **10**. The battery cover **110** and the middle frame **401** form an accommodation space. The main board **10**, the battery **30**, and another electronic component are accommodated in the accommodation space. The another electronic component may be a camera, a key, a speaker, or the like. A material of the middle frame **401** may be a metal material having specific hardness, such as aluminum alloy.

(20) In this disclosure, the heat pipe **60** and the first vapor chamber **70** are integrated on the middle frame **401** to form the middle frame assembly **40**, to improve heat dissipation performance of the mobile terminal. The vapor chamber and the heat pipe each have a vacuum heat conducting cavity. An inner wall of the heat conducting cavity has a capillary structure, and working fluid is injected, where the working fluid may be pure water. Working principles of two-phase heat dissipation parts such as the VC and the heat pipe are similar, and include four main steps: conduction, evaporation, convection, and coagulation. The vapor chamber is used as an example to provide the following description. Heat generated by a heat source enters the vapor chamber through heat conduction, and working fluid near the heat source absorbs the heat and then rapidly vaporizes the heat while taking away a large amount of heat. Then, with a latent heat property of the vapor, when the vapor in the chamber is diffused from a high-pressure region (namely, a high-temperature region) to a low-pressure region (namely, a low-temperature region), and the vapor is in contact with the inner wall with a relatively low temperature, the vapor rapidly condenses into a liquid state and releases heat. The working fluid condensed into the liquid state returns to the heat source under action of capillary force of the capillary structure, to complete one time of heat conduction circulation, and form a two-way circulation system in which the working fluid has both a vapor phase and a liquid phase. An area of the vapor chamber is larger than that of the heat pipe, so that two-dimensional extended heat conduction can be implemented. A width of the heat pipe is relatively small, and one-dimensional linear heat conduction can be performed. Specifically, one-dimensional heat conduction is performed along an extension direction of the heat pipe. A material of the heat pipe is usually a metal material, such as aluminum or copper. A material of the vapor chamber is also usually metal, for example, copper may be selected.

(21) In this disclosure, the heat pipe **60** and the vapor chamber (such as the first vapor chamber **70**) may have a same physical property parameter, or may be different in at least one physical property parameter, for example, different housing material combinations, different internal working fluid materials, different capillary structures (cross-sectional areas of capillary layers), or the like; or one or more of an amount of working fluid (namely, an amount of filled working fluid), a type of working fluid, and a pipe material (which may be copper or aluminum) and a thickness that are of the heat pipe **60**.

(22) FIG. 2 is a schematic top view of a structure of the middle frame **401** according to an embodiment of this disclosure. The middle frame **401** includes a top **401a**, a bottom **401b** disposed opposite to the top **401a**, and two sidewalls **401c** and **401d** that are disposed opposite to each other between the top **401a** and the bottom **401b**. The top **401a**, the bottom **401b**, and the two sidewalls **401c** and **401d** described herein refer to four sides on a periphery of the middle frame (which are four sides in the schematic diagram shown in FIG. 2, and are actually four outer surfaces of the middle frame from a three-dimensional perspective, where the four surfaces are surfaces of a bezel connected between a screen and a rear cover of the mobile terminal). For example, the mobile

terminal is a mobile phone. The middle frame is basically rectangular, and the two sidewalls **401c** and **401d** are long sides of the middle frame. The middle frame **401** may include a heat dissipation region **402** corresponding to a heat source **101**, a battery placement region **403** corresponding to the battery **30**, and another element placement region **404**. The heat dissipation region **402** is located between the battery placement region **403** and the top **401a**, and the another element placement region **404** is located between the battery placement region **403** and the bottom **401b**. In another implementation, the heat dissipation region **402** may be located between the battery placement region **403** and one sidewall.

(23) The heat dissipation region **402** is used to provide partial space to install one or more heat dissipation components (such as the VC or the heat pipe) to dissipate heat from the heat source. For example, the heat dissipation region **402** may be a removed region on the middle frame **401**, that is, a region in which a part of the middle frame material is removed. Specifically, a part of the middle frame material may be removed to form a “groove”, or all of the middle frame material may be removed to form a “hole”. Certainly, in some other embodiments, a material in the heat dissipation region **402** may not be removed. To implement a thin-type design of the mobile terminal, a hole may be dug on the middle frame **401**, and a region in which the hole is dug may be used to accommodate a heat source (such as a processor) on the main board, and then a VC is installed on the heat source to dissipate heat from the heat source. The heat dissipation region **402** may be a solid region on the middle frame **401**, and a hole is not dug. However, a groove may be dug to accommodate the heat source or the VC (for example, a groove is dug on both a front surface and a back surface to accommodate both the VC and the heat source, or a groove is dug on one surface to accommodate the VC or the heat source). In another embodiment, neither a hole nor a groove is dug, or both a hole and a groove may be dug (both a groove and a hole may be provided in one region, or a groove and a hole are disposed at intervals). Similarly, the battery placement region **403** and the another element placement region **404** may be hollow regions or solid regions on the middle frame **401** or a combination of the two regions. The “groove” in this disclosure refers to a concave structure that does not penetrate the middle frame, and a shape is not limited (for example, may be a rectangle, a square, a circle, or various irregular shapes). The “hole” (which may also be understood as a “hole” or an “opening”) in this disclosure is a structure formed by penetrating the middle frame, and a shape is also not limited.

(24) The battery placement region **403** includes a first edge **403a**, a second edge **403b**, a third edge **403c**, and a fourth edge **403d**. The heat dissipation region **402** is located between the first edge **403a** of the battery placement region **403** and the top **401a**. The another element placement region **404** is disposed between the second edge **403b** of the battery placement region **403** and the bottom **401b**, and a gap **50** is disposed between the battery placement region **403** and the sidewall **401c** of the middle frame **401**. Generally, no electronic component is disposed at a location between the battery placement region **403** and the sidewall **401c** of the middle frame **401**. In this disclosure, the heat pipe **60** is disposed at a location of the gap **50**, so that heat dissipation performance can be improved.

(25) Specifically, a first gap **501** is disposed between the third edge **403c** and the first sidewall **401c**, and a second gap **502** is disposed between the fourth edge **403d** and the second sidewall **401d**. The middle frame **401** at locations of the first gap **501** and the second gap **502** may be provided with a groove or a hole. The hole refers to a through-hole shape dug on the middle frame through excavation. The groove and the hole may be disposed to implement a thin-type design of the mobile terminal or reduce a weight of the middle frame **401**. However, to ensure rigidity of the middle frame **401**, the groove or the hole may be disposed at a proper location, and a thickness may be reserved at a proper location to ensure overall rigidity of the middle frame **401**, so that the middle frame **401** is not easily deformed and can bear an electronic component.

(26) FIG. 3 is a schematic top view of a structure of the middle frame assembly **40**. The first vapor chamber **70** is accommodated in the heat dissipation region **402**. Specifically, the first vapor

chamber **70** may be fixedly connected to the middle frame **401** in the heat dissipation region **402**. A specific fastening manner is not limited to the following several manners: The first vapor chamber **70** may be fastened to the middle frame **401** through bottom welding, may be fastened to the middle frame **401** through side welding, may be fastened to the middle frame **401** through bottom bonding, may be fixedly connected to the middle frame **401** by using circumferential overlapping structures, or may be connected to the middle frame **401** through combination of bottom surface fastening and circumferential fastening. The heat pipe **60** is connected to the first vapor chamber **70** and is configured to dissipate heat from the first vapor chamber **70**. A manner of connecting the heat pipe **60** and the first vapor chamber **70** is not limited to fastening through overlapping and welding, sealing through welding after the heat pipe **60** and the first vapor chamber **70** are connected, or the like. A thickness of the first vapor chamber **70** may be less than or equal to 0.8 mm, and a nominal pipe diameter of the heat pipe **60** may be less than or equal to 8 mm. To make the mobile terminal thinner, a thickness of the heat pipe **60** may be less than or equal to 0.6 mm. A small-sized heat pipe **60** may extend between electronic components or between the electronic component and the sidewall of the middle frame **401**, to use space between the electronic components and a gap between the electronic component and the sidewall of the middle frame **401**, thereby facilitating a miniaturized design of the mobile terminal while improving heat dissipation performance. The thickness of the first vapor chamber **70** may be the same as or different from the thickness of the heat pipe **60**. The heat pipe **60** may be manufactured to have an unequal thickness, an unequal width, a bend, a section difference, or the like, which is determined based on a specific use scenario.

(27) In this disclosure, one end of the heat pipe **60** is connected to the first vapor chamber **70**, and heat of the heat source **101** may be rapidly dissipated by using a comprehensive heat conduction effect of two-dimensional extended heat conduction of the vapor chamber and one-dimensional linear heat conduction of the heat pipe **60**, to prolong a service life of the heat source **101** and reduce a temperature of the heat source, thereby prolonging a service life of the mobile terminal, reducing a temperature of the mobile terminal during use, and improving use experience of a consumer. In addition, the first vapor chamber **70** may be fastened to the middle frame **401**. For example, the first vapor chamber **70** may be fixedly connected to the middle frame **401** through welding. The heat of the heat source **101** may be conducted to the entire middle frame **401** by using the first vapor chamber **70**, and the heat is conducted out by using the middle frame **401**. An area of the middle frame assembly **40** is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal.

(28) In this disclosure, a manner of fastening the first vapor chamber **70** in the heat dissipation region **402** includes but is not limited to the following two manners.

(29) Manner 1: The heat dissipation region **402** is a hole. In this way, even if the thickness of the first vapor chamber **70** is relatively large, the hole can still accommodate the relatively thick first vapor chamber **70**, and the first vapor chamber **70** does not increase a thickness of the middle frame assembly **40**. In addition, the hole is disposed to prevent, to the utmost extent, the first vapor chamber **70** from increasing the thickness of the middle frame assembly **40**. Moreover, to implement better heat dissipation and fastening, the first vapor chamber **70** may be further fixedly connected to the middle frame **401** by using an overlapping structure. Specifically, the middle frame **401** includes a first overlapping side **401a**, and the first overlapping side **401a** extends toward the inside of the hole. The first vapor chamber **70** includes a third overlapping side **70a**, and the third overlapping side **70a** is disposed protruding from a periphery of the first vapor chamber **70**. The first overlapping side **401a** overlaps the third overlapping side **70a**. The first overlapping side **401a** and the third overlapping side **70a** may be fastened through welding or may be fastened by using an adhesive at a location at which they overlap. In this way, the first vapor chamber **70** is fastened to the middle frame **401** (refer to FIG. **1b**). The first vapor chamber is further configured to connect to a heat source to dissipate heat from the heat source. For example, the first vapor

chamber is connected to the heat source and dissipate heat by using a heat conducting adhesive.

(30) Manner 2: The heat dissipation region **402** is a groove. To be specific, a part of the middle frame material of a specific thickness is further reserved in the heat dissipation region **402** to bear the first vapor chamber **70**. The first vapor chamber **70** is embedded in the groove, and the first vapor chamber **70** may be fastened and adhered to a bottom wall of the groove by using a heat conducting adhesive, that is, the first vapor chamber **70** and the middle frame **401** are fastened by using the heat conducting adhesive. This fastening architecture requires a relatively thick heat conducting adhesive. In another implementation, a relatively thin layer of heat conducting adhesive may be disposed. In addition, an overlapping structure is disposed on the middle frame **401**, an overlapping structure is disposed at an edge of the first vapor chamber **70**, and the overlapping structure of the middle frame **401** (for details, refer to the overlapping structure in the first fastening manner) cooperates with the overlapping structure of the first vapor chamber to form an overlapping fastening solution. This overlapping fastening solution and the heat conducting adhesive jointly fasten the first vapor chamber **70** to the middle frame **401**. In this manner, the first vapor chamber **70** is embedded in the groove, so that the first vapor chamber **70** does not increase a thickness of the middle frame assembly **40**, the middle frame assembly **40** may be relatively thin, and the mobile terminal may be relatively thin. It may be understood that the heat dissipation region **402** is further configured to connect to a heat source to dissipate heat from the heat source. For example, the heat dissipation region **402** is connected to the heat source by using a heat conducting adhesive, to first conduct, to the heat dissipation region **402** in the middle frame, heat generated by the heat source, and then the heat is dissipated by using the first vapor chamber fastened in the heat dissipation region **402**.

(31) In this disclosure, there is at least one heat pipe **60**, and there is at least one first vapor chamber **70**. One or more heat pipes **60** or first vapor chambers **70** may be selected based on an actual engineering design requirement (such as an internal space structure layout and a heat dissipation requirement) or the like. For example, in the implementation shown in FIG. **3**, there is one heat pipe **60** and one vapor chamber (namely, the first vapor chamber **70**). It may be understood that when there are a plurality of heat pipes **60** or a plurality of vapor chambers, a heat dissipation area of the middle frame assembly **40** may be enlarged, and heat dissipation efficiency is high. In one of the following embodiments, one heat pipe **60** and one vapor chamber (the first vapor chamber **70**) are described.

(32) A connection relationship between the heat pipe **60** and the first vapor chamber **70** is described below in detail.

(33) Refer to FIG. **3**. The middle frame assembly **40** shown in FIG. **3** includes the middle frame **401**, one heat pipe **60**, and one vapor chamber (the first vapor chamber **70**). The heat pipe **60** includes a first section **60a** and a second section **60b** disposed opposite to the first section **60a**. The first section **60a** and the second section **60b** each refer to a segment of solid region in which the heat pipe **60** extends from an end to a direction opposite to the end. Specifically, in the embodiment shown in FIG. **3**, the heat pipe includes two ends. The first section **60a** is connected to the first vapor chamber **70**. The second section **60b** is a free end and may be completely or partially connected to a component (for example, the middle frame, the battery, or another electronic component) that is cold relative to the first vapor chamber. In another embodiment, the heat pipe **60** may have a plurality of branches, and therefore has a plurality of ends. In this case, there may be more sections. Some sections may be connected to the first vapor chamber **70**, and the other sections may be connected to a component that is cold relative to the first vapor chamber.

(34) After the heat source **101** runs and generates heat, the heat may be first transferred to the first vapor chamber **70** (assuming that the heat dissipation region **402** is a hole, the first vapor chamber **70** passes through the hole to be connected to the heat source, or the first vapor chamber **70** is accommodated in the hole to be connected to the heat source). The heat on the first vapor chamber **70** may be transferred to the second section **60b** along the first section **60a** of the heat pipe **60**. The

second section **60b** is relatively far away from the heat source **101**, and the second section **60b** may rapidly dissipate the heat. Further, the heat of the heat source **101** may be conducted to the entire heat pipe **60**, and heat conduction circulation may be completed in the entire heat pipe **60**. Therefore, a heat dissipation area is large, and heat dissipation efficiency is high, thereby improving heat dissipation performance of the entire mobile terminal. It may be understood that when the heat source **101** generates a relatively small amount of heat, the heat may be transferred to a specific location between the first section **60a** and the second section **60b**, to complete one time of heat conduction circulation. In addition, when the first vapor chamber **70** is fastened to the middle frame **401** in some heat conducting manners (such as welding or bonding by using a heat conducting adhesive), the heat on the first vapor chamber **70** may be further transferred to the entire middle frame **401**, and the heat is further dissipated by using the middle frame **401**.

(35) In another implementation, both the heat pipe **60** and the vapor chamber (such as the first vapor chamber **70**) may be fastened to the middle frame **401** (as shown in FIG. **1b**). Heat generated by the heat source **101** during running may be first transferred to the first section **60a** of the heat pipe **60**, and the first section **60a** of the heat pipe **60** transfers the heat to the first vapor chamber **70** and transfers the heat to the second section **60b** of the heat pipe **60**. In another implementation, the heat source **101** may include electronic components of different heights. Therefore, a relatively high electronic component may be directly in contact with the first vapor chamber **70**, and a gap may be formed between a relatively small-sized electronic component and the first vapor chamber **70**. The heat pipe **60** may be disposed at a location of the gap. Both the first section **60a** of the heat pipe **60** and the first vapor chamber **70** may be in contact with the heat source **101**, so that the heat generated by the heat source **101** during running may be transferred to both the first section **60a** of the heat pipe and the first vapor chamber **70**. In this way, the heat pipe **60** may be disposed in relatively small space, to transfer heat to a region in which the mobile terminal generates no heat. In addition, two-dimensional heat conduction of the first vapor chamber **70** may be further used, so that a heat conduction effect is fairly good, and the heat is transferred to a region with a relatively low temperature in the mobile terminal by using the second section **60b** of the heat pipe **60**.

(36) In this disclosure, the heat pipe **60** is combined with the first vapor chamber **70**, and the heat is transferred to the region with a relatively low temperature in the mobile terminal by using the second section **60b** of the heat pipe, to implement efficient heat dissipation. The heat pipe **60** extends from the heat dissipation region **402** to a direction of the second edge **403b**. In other words, the heat pipe **60** extends along a direction from the first edge **403a** to the second edge **403b** of the battery placement region **403**, and the second section **60b** of the heat pipe **60** may be located between the first edge **403a** and the second edge **403b** (as shown in FIG. **3**). It may be understood that the second section **60b** of the heat pipe **60** may cross the second edge **403b** and is located in the another element placement region **404** (as shown in FIG. **4**). It may be understood that the gap between the battery placement region **403** and the sidewall **401c** of the middle frame **401** and the another element placement region **404** both may be considered as regions with a relatively low temperature, and temperatures in the two places both are lower than a temperature in the heat dissipation region **402**.

(37) Still refer to FIG. **3**. FIG. **3** shows that the heat pipe **60** extends along a periphery of the battery placement region **403**. The periphery of the battery placement region **403** may be a region between the battery placement region **403** and the sidewall of the middle frame **401**, or may be a region between the battery placement region **403** and the heat dissipation region **402**. When the heat pipe **60** extends along the periphery of the battery placement region **403**, the heat pipe **60** does not overlap the battery placement region **403**, and does not affect a thickness of the middle frame assembly **40**. When the battery is placed in the mobile terminal, the heat pipe **60** is not in contact with the battery, and does not affect a thickness of the mobile terminal.

(38) Specifically, the heat pipe **60** is connected to a part that is in the middle frame **401** and whose temperature is lower than a temperature of the first vapor chamber **70** but is not in contact with the

battery. To be specific, when the mobile terminal runs normally, the mobile terminal generates heat. The temperature of the part that is in the middle frame **401** and that is connected to the heat pipe **60** is less than that of the first vapor chamber **70**. The part that is in the middle frame **401** and whose temperature is less than that of the first vapor chamber **70** may be a part on a side that is of the middle frame **401** and that is away from the first vapor chamber **70**. In other words, the heat pipe **60** is further connected to the side that is of the middle frame **401** and that is away from the first vapor chamber **70**. A heat dissipation area of the middle frame **401** is large, and heat dissipation efficiency is high. In addition, when the heat pipe **60** is not in contact with the battery, the heat pipe **60** and the battery do not transfer heat to each other. The heat pipe **60** does not overlap the battery and does not affect the thickness of the mobile terminal. It may be understood that the first vapor chamber **70** is configured to dissipate heat from the heat source **101**, for example, dissipate heat from a CPU. A part that is in the middle frame assembly **40** and whose temperature is lower than the temperature of the first vapor chamber **70** may be located in a bottom region of the mobile terminal, or may be located in a region near a boundary of the middle frame assembly **40**. An antenna, a key, a speaker, or the like may be placed in these regions. When the mobile terminal runs normally, temperatures in these regions are lower than the temperature of the first vapor chamber **70**, and the heat pipe **60** may be connected to these regions or may be connected to components in these regions.

(39) In one embodiment, the gap **50** between the battery placement region **403** and the sidewall **401c** of the middle frame **401** is idle space, and a part of the heat pipe **60** may be accommodated in the gap **50** between the battery placement region **403** and the sidewall **401c** of the middle frame **401**, that is, a part of the heat pipe **60** is accommodated in the idle space. In this way, not only the idle space is used, but also no new space needs to be disposed in the mobile terminal to accommodate the heat pipe **60**, so that a volume of the mobile terminal is not increased. In another implementation, a relatively large quantity of gaps are disposed between the battery placement region **403** and the top **401a** of the middle frame **401**, for example, a gap between electronic components. A part of the heat pipe **60** may be accommodated in the gap between the electronic components and extends in the gap. The gap may be located on a side that is of the middle frame **401** and that corresponds to the main board **10**.

(40) FIG. 5 and FIG. 6 show connection relationships between the heat pipe **60** and the first vapor chamber **70**. The connection relationships between the heat pipe **60** and the first vapor chamber **70** include but are not limited to the following three types.

(41) Type 1: As shown in FIG. 5, the heat pipe **60** overlaps the first vapor chamber **70** in a direction perpendicular to a chamber surface of the first vapor chamber **70**. Specifically, the first section **60a** overlaps the first vapor chamber **70** in the direction perpendicular to the first vapor chamber **70**, to be specific, the first section **60a** is located outside the first vapor chamber **70** and partially overlaps the first vapor chamber **70**. The first section **60a** may be connected to the first vapor chamber **70** by using a heat conducting adhesive, to improve heat conduction performance. Through overlapping between the heat pipe **60** and the first vapor chamber **70**, heat transferred from the heat source **101** to the first vapor chamber **70** may be transferred to the heat pipe **60** through heat conduction, so that not only heat can be transferred rapidly, but also a manner of connecting the heat pipe **60** and the first vapor chamber **70** is simple and easy to operate. In a specific implementation, same working fluid may be disposed in the heat pipe **60** and the first vapor chamber **70**. Certainly, different working fluid may be disposed in the heat pipe **60** and the first vapor chamber **70**. For example, first working fluid may be disposed in the heat pipe **60**, and second working fluid may be disposed in the first vapor chamber **70**, where the first working fluid is different from the second working fluid. For example, the first working fluid may be methanol, R134A, acetone, or the like, and the second working fluid may be water or the like. The heat pipe **60** and the first vapor chamber **70** may be connected by using a heat conducting medium. The heat conducting medium may be a heat conducting adhesive or a graphite material. Heat conductivity of the heat conducting

adhesive may be more than 1 W/m.K. The heat conductivity of the heat conducting adhesive is twice over heat conductivity of an ordinary adhesive layer, and the heat conducting adhesive has good heat conduction performance.

(42) Type 2: As shown in FIG. 6, the heat pipe **60** is connected to the first vapor chamber **70**. Specifically, a first cavity **7** is disposed in the first vapor chamber **70**, a second cavity **6** is disposed in the heat pipe **60**, and the first section **60a** extends into the first vapor chamber **70**. A hole may be dug on the first vapor chamber **70**, and the first section **60a** of the heat pipe **60** extends into the hole and is fastened through welding and sealing. In the solution in which the first section **60a** extends into the first vapor chamber **70**, internal space of the first vapor chamber **70** may be connected to internal space of the heat pipe **60**, to form a relatively large and complete heat conducting cavity. As shown in FIG. 6, a relatively large and complete heat conducting cavity is formed inside after the heat pipe and the first vapor chamber are connected. A capillary structure a in the first vapor chamber **70** is connected to a capillary structure b in the heat pipe **60**, and working fluid in the heat conducting cavity passes through a longer path and takes longer time to complete one time of heat conduction circulation, so that heat of the heat source can be transferred to a larger region of the middle frame, and the heat is conducted out more dispersedly, and therefore heat dissipation efficiency is high. It may be understood that a state in which the first vapor chamber **70** is connected to the heat pipe **60** is equivalent to extending a conduction path of a heat conducting medium in the heat conducting cavity, and a temperature may be equalized between a region in which heat sources are centralized and a region with a relatively low temperature in the terminal. This architecture has a better temperature equalization effect.

(43) Type 3: The first section **60a** is connected to a side of the first vapor chamber **70**. The first vapor chamber **70** and the first section **60a** may be fastened through welding by using another connecting piece, or the first section **60a** may be fixedly connected to an overlapping side disposed at an outer edge of the first vapor chamber **70**.

(44) In this disclosure, electronic components in the mobile terminal may have different heights. For example, components forming the heat source **101** have different heights, so that a partial region of the middle frame **401** is relatively thick and a partial region of the middle frame **401** is relatively thin. Therefore, a thickness of the vapor chamber in this disclosure may also be uneven. For example, refer to FIG. 7. The first vapor chamber **70** includes three parts of different thicknesses. The first cavity **7** in the first vapor chamber **70** is also distributed with different volumes. The capillary structure a is disposed in the first cavity **7**, and the capillary structure a is distributed in a step shape. A plurality of support posts are further distributed in the first cavity **7** of the first vapor chamber **70**, which are configured to support the first cavity **7**, to prevent the first vapor chamber **70** from being deformed, thereby reducing internal space of the first cavity. In this implementation, a thickness of the first vapor chamber **70** is increased step by step, which may be applied to an electronic component whose height is decreased step by step. This further enables a volume of a flow equalization cavity inside the first vapor chamber **70** to gradually increase, and enables a flowing range of working fluid to be larger, so that a heat dissipation effect is better. It may be understood that, the thickness of the first vapor chamber **70** may be uneven in another manner. For example, the thickness of the first vapor chamber **70** is evenly alternating between high and low or unevenly alternating between high and low, to correspond to electronic components of different heights; or the thickness of the first vapor chamber **70** may be linearly decreased or linearly increased. The thickness of the first vapor chamber **70** in this disclosure is determined based on a height of an electronic component in a specific use environment.

(45) In this disclosure, two-phase components (the heat pipe **60** and the vapor chamber) of different thicknesses may be used in different regions, and two-phase components (the heat pipe **60** and the vapor chamber) of different widths may be used in different regions, to flexibly arrange the two-phase components (the heat pipe **60** and the vapor chamber) on the middle frame **401** of the mobile terminal.

(46) FIG. 8 is another schematic top view of a structure of the middle frame assembly 40, and the figure shows that the middle frame assembly 40 includes one heat pipe 60 and two vapor chambers (the first vapor chamber 70 and a second vapor chamber 80). The second vapor chamber 80 is fastened in the another element placement region 404, and the heat pipe 60 is connected to the first vapor chamber 70 and the second vapor chamber 80. In this embodiment, the first section 60a of the heat pipe 60 is fixedly connected to the first vapor chamber 70, and the second section 60b of the heat pipe 60 is fixedly connected to the second vapor chamber 80. To be specific, when the heat source is fixedly placed in the heat dissipation region 402, and another heat source is fixedly placed in the another element placement region 404, heat of the heat source may be conducted to the first vapor chamber 70 and conducted to the heat pipe 60, and heat of the another heat source may be conducted to the second vapor chamber 80 and conducted to the heat pipe 60. Further, the heat of the heat source and the another heat source may be rapidly dissipated under comprehensive action of the first vapor chamber 70, the second vapor chamber 80, and the heat pipe 60, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer.

(47) FIG. 9 shows a manner of connecting the heat pipe 60 to the first vapor chamber 70 and overlapping the heat pipe 60 with the second vapor chamber 80. The first vapor chamber 70 includes a first top surface 701 and a first bottom surface 702 disposed opposite to the first top surface 701, and the second vapor chamber 80 includes a second top surface 801 and a second bottom surface 802 disposed opposite to the second top surface 801. A manner of connecting the heat pipe 60 to the first vapor chamber 70 and the second vapor chamber 80 includes but is not limited to the following four manners: The heat pipe 60 is connected to the first top surface 701 and the second top surface 801. The heat pipe 60 is bent to be connected to the first top surface 701 and the second bottom surface 802. The heat pipe 60 is bent to be connected to the first bottom surface 702 and the second top surface 801 (FIG. 9). The heat pipe 60 is connected to the first bottom surface 702 and the second bottom surface 802. All the foregoing four manners of connecting the heat pipe 60 to the first vapor chamber 70 and the second vapor chamber 80 can be used to conduct heat of the heat source 101 and another heat source 102 to the first vapor chamber 70, the second vapor chamber 80, and the heat pipe 60, and conduct the heat to the entire middle frame 401. A specific manner to be selected is determined based on an actual use requirement. Working fluid in the heat pipe 60, the first vapor chamber 70, and the second vapor chamber 80 may be the same or may be different. For example, first working fluid is disposed in the heat pipe 60, second working fluid is disposed in the first vapor chamber 70, and third working fluid is disposed in the second vapor chamber 80. At least one of the first working fluid, the second working fluid, and the third working fluid is different from remaining working fluid, and the first working fluid, the second working fluid, and the third working fluid may be one, two, or three of methanol, R134A, acetone, or water.

(48) FIG. 10 shows a manner of connecting the heat pipe 60 to the first vapor chamber 70 and the second vapor chamber 80. Internal space of the heat pipe 60, internal space of the first vapor chamber 70, and internal space of the second vapor chamber 80 jointly form a heat conducting cavity. Specifically, the heat pipe 60 may be connected to one of the first vapor chamber 70 and the second vapor chamber 80, or the heat pipe 60 is connected to both the first vapor chamber 70 and the second vapor chamber 80. A connection manner is the same as the foregoing manner of connecting the heat pipe 60 to the first vapor chamber 70, and details are not described herein again in this disclosure. A larger and complete heat conducting cavity is formed inside after the heat pipe 60, the first vapor chamber 70, and the second vapor chamber 80 are connected. Working fluid in the heat conducting cavity passes through a longer path and takes longer time to complete one time of heat conduction circulation, so that heat of the heat source may be transferred to a larger region of the middle frame 401, and the heat is conducted out more dispersedly, and therefore heat dissipation efficiency is high.

(49) A manner of fastening the second vapor chamber 80 to the middle frame 401 in the another

element placement region **404** may be the same as a manner of fastening the first vapor chamber to the middle frame **401** in the heat dissipation region **402**. The middle frame **401** may be provided with a first groove, a second groove, and a third groove. The first groove is disposed in the heat dissipation region **402**, the second groove is disposed in the another element placement region **404**, the first vapor chamber **70** is embedded in the first groove, the second vapor chamber **80** is embedded in the second groove, and the heat pipe **60** is disposed in the third groove. In this manner, the vapor chamber and the heat pipe are embedded in the grooves, so that the first vapor chamber **70**, the second vapor chamber **80**, and the heat pipe **60** do not increase a thickness of the middle frame **401**, and the mobile terminal may be relatively thin. It may be understood that the first groove, the second groove, and the third groove may be simultaneously disposed in one implementation. In another implementation, one or two of the three grooves may be selectively configured based on a specific structure in the mobile terminal. For example, in some places, a gap exists between electronic components and can be used for placing the heat pipe. In this case, the third groove does not need to be disposed.

(50) Certainly, it may be understood that the middle frame **401** is provided with the first groove and the third groove, and the third groove is formed at the bottom of the first groove. The first vapor chamber **70** is embedded in the first groove, the heat pipe **60** is partially embedded in the third groove, and a part of the heat pipe **60** in the groove is stacked with the first vapor chamber **70**. In this implementation, a part of the heat pipe **60** is located between the first vapor chamber **70** and the heat source **101**, openings of the first groove and the third groove face a side that is of the middle frame **401** and that is away from the heat source **101**, the heat source **101** is spaced from the heat pipe **60** and the first vapor chamber **70** by the middle frame **401**, and the middle frame **401** is configured to support the heat source **101**. In this way, the heat of the heat source **101** is first transferred to the middle frame **401** opposite to the heat source **101** and is then transferred to other parts of the first vapor chamber **70**, the heat pipe **60**, and the middle frame **401**, so that heat of the mobile terminal can be rapidly dissipated.

(51) In some other embodiments, a “hole” may be used to replace one or more of the “grooves”, that is, the first groove, the second groove, and the third groove each may be in a form of the “hole”. Alternatively, in the three grooves, some may be designed as grooves, and some may be designed as architectures of holes.

(52) Refer to FIG. **11**. A wire groove **608** is disposed on an outer surface of the heat pipe **60**. The wire groove **608** is configured to fasten a conducting wire in the mobile terminal, and the conducting wire is configured to electrically connect an electronic component in the heat dissipation region **402** and an electronic component in the another element placement region. Specifically, for example, the heat source may be a plurality of high-power components such as a CPU or a GPU, and the another heat source may be a heat source such as sound on display and a screen driver. The foregoing heat source needs to be connected to the another heat source by using a conducting wire. The wire groove **608** on the surface of the heat pipe **60** may be used as a carrier of the conducting wire, so that the conducting wire may be fastened in the wire groove **608** of the heat pipe **60**, to fasten a location of the conducting wire, thereby avoiding damage caused to the conducting wire due to shaking, and prolonging a service life of the mobile terminal. A specific manner of disposing the wire groove **608** on the surface of the heat pipe **60** may be as follows: A groove is formed on the outer surface of the heat pipe **60** through processing by a machine; or a structure, such as a hook similar to a wire bundle, used to fasten a conducting wire is welded on the outer surface of the heat pipe **60**, and the wire groove **608** may be formed in the hook.

(53) In a specific embodiment, a heat conducting film is disposed on outer surfaces of the heat pipe **60** and the vapor chambers (the first vapor chamber **70** and the second vapor chamber **80**). A material of the heat conducting film may be graphene. The heat conducting film is disposed, so that heat on the heat pipe and the vapor chamber is conducted to the outside more rapidly.

(54) A specific implementation in which the middle frame assembly **40** includes two heat pipes **60**

and one vapor chamber (the first vapor chamber **70**) is described below.

(55) FIG. **12** is another schematic top view of a structure of the middle frame assembly **40**. In this embodiment, the heat pipe **60** includes at least a first heat pipe **603** and a second heat pipe **604** that are disposed at intervals, and both the first heat pipe **603** and the second heat pipe **604** are disposed around the battery placement region **403**. Disposing of the two heat pipes **60** achieves a good heat dissipation effect, and heat of the heat source can be dissipated more rapidly, to prolong a service life of the heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, when both the first heat pipe **603** and the second heat pipe **604** are disposed around the battery placement region **403**, a gap between the battery placement region **403** and each of the third sidewall **401c** and the fourth sidewall **401d** is used, and neither of the two heat pipes **60** overlaps the battery placement region **403** and affects a thickness of the middle frame **401**. When the battery is placed in the mobile terminal, the heat pipe **60** does not overlap the battery, and does not affect the thickness of the mobile terminal.

(56) In a specific implementation, the first section **60a** of the first heat pipe **603** is accommodated between the top **401a** and the battery placement region **403** and is near the third edge **403c**. The second section **60b** of the first heat pipe **603** extends to the first gap **501** between the third edge **403c** and the third sidewall **401c**. A part of the second heat pipe **604** is accommodated between the top **401a** and the battery placement region **403** and is near the fourth edge **403d**. A part of the second heat pipe **604** extends to the second gap **502** between the fourth edge **403d** and the fourth sidewall **401d**.

(57) FIG. **13** is another schematic top view of a structure of the middle frame **401**. There are still two heat pipes **60** shown in the figure. The second heat pipe **604** surrounds the battery placement region **403**, and the first heat pipe **603** overlaps the battery placement region **403**. When space between the battery placement region **403** and the sidewall **401c** of the middle frame **401** is insufficient to accommodate the two heat pipes **60**, one of the heat pipes **60** may overlap the battery placement region **403**. In this way, heat dissipation effects of the two heat pipes **60** can be used, and the middle frame **401** has a larger region for heat dissipation, so that a heat dissipation effect is better, and heat of the heat source can be dissipated more rapidly, to prolong a service life of the heat source, thereby prolonging a service life of the mobile terminal and improving use experience of a consumer. In addition, space between the battery placement region **403** and each of the third sidewall **401c** and the fourth sidewall **401d** of the middle frame **401** and other space can also be effectively used. Moreover, the manner in which the first heat pipe **603** overlaps the battery placement region **403** facilitates installation of the first heat pipe **603** and is convenient for operation, and the first heat pipe **603** may be configured to dissipate heat of the battery.

(58) Specifically, when the first heat pipe **603** overlaps the battery placement region **403**, the first heat pipe **603** is connected to a surface of the battery, so that heat on the battery may be further transferred to the first heat pipe **603** and is then transferred to the entire middle frame **401**. In this way, the middle frame **401** dissipates the heat from the battery. A heat dissipation area is large, and a heat dissipation effect is fairly good.

(59) In another embodiment, the middle frame assembly **40** may include two heat pipes **60** and two vapor chambers, such as the first heat pipe **603**, the second heat pipe **604**, the first vapor chamber **70**, and the second vapor chamber **80**. The first vapor chamber **70** is fastened in the heat dissipation region **402**, the second vapor chamber **80** is fastened in the another element placement region **404**, and both the first heat pipe **603** and the second heat pipe **604** surround the battery placement region **403**. The first section **60a** of the first heat pipe **603** is connected to the first vapor chamber **70**, the second section **60b** of the first heat pipe **603** is connected to the second vapor chamber **80**, the first section **60a** of the second heat pipe **604** is connected to the first vapor chamber **70**, and the second section **60b** of the second heat pipe **604** is connected to the second vapor chamber **80**.

(60) Certainly, in another implementation, there may be more heat pipes **60** and vapor chambers, such as a third vapor chamber and a third heat pipe **60**. More vapor chambers are connected to

more heat pipe **60** to implement heat dissipation in a larger area.

(61) At least one of the one or more heat pipes **60** provided in this disclosure extends between electronic components, and/or at least one of the one or more heat pipes **60** extends between the electronic component and a boundary of the middle frame **401**. The electronic component may be a backlight module, a battery, a CPU, a camera, a key, a speaker, or the like in a terminal. As shown in FIG. **13**, the first heat pipe **603** extending between electronic components may be understood as the first heat pipe **603** extending between the battery and the backlight module stacked with the battery. The second heat pipe **604** extending between the electronic component and the boundary of the middle frame may be understood as that the second heat pipe **604** extending between the battery and the boundary of the middle frame. According to the middle frame assembly **40** and the mobile terminal in this disclosure, a comprehensive heat conduction effect of the heat pipe **60** and the vapor chamber is used, thereby improving a heat dissipation effect and improving use experience of a consumer.

(62) What are disclosed above are merely example embodiments of this disclosure, and certainly are not intended to limit the protection scope of this application. A person of ordinary skill in the art may understand that all or some of processes that implement the foregoing embodiments and equivalent modifications made in accordance with the claims of the present invention shall fall within the scope of this disclosure.

Claims

1. A mobile terminal comprising a mainboard, a heat source and a middle frame assembly, wherein: the mainboard is installed on the middle frame assembly, and the heat source is disposed on the mainboard; the middle frame assembly comprises a middle frame, one or more heat pipes, and a first vapor chamber; the middle frame comprises a heat dissipation region corresponding to the heat source, and the first vapor chamber is accommodated in the heat dissipation region and is configured to dissipate heat from the heat source; the one or more heat pipes are connected to the first vapor chamber and are configured to dissipate heat from the first vapor chamber; the mobile terminal includes electronic components and an internal conducting wire; and a wire groove is disposed on an outer surface of at least one of the one or more heat pipes, the wire groove is configured to fasten the internal conducting wire, and the conducting wire is configured to electrically connect the electronic components.
2. The mobile terminal according to claim 1, wherein at least one of the one or more heat pipes is further connected to a part that is in the middle frame and whose temperature is lower than a temperature of the first vapor chamber.
3. The mobile terminal according to claim 2, wherein at least one of the one or more heat pipes is connected to the middle frame by using a heat conducting medium.
4. The mobile terminal according to claim 1, wherein the one or more heat pipes are connected to the first vapor chamber by using a heat conducting medium.
5. The mobile terminal according to claim 1, wherein an internal space of at least one of the one or more heat pipes is connected to an internal space of the first vapor chamber to form a heat conducting cavity.
6. The mobile terminal according to claim 5, wherein one end of at least one of the one or more heat pipes is inserted into the first vapor chamber.
7. The mobile terminal according to claim 1, wherein the mobile terminal further comprises a battery, the middle frame further comprises a battery placement region corresponding to the battery, the battery placement region comprises a first edge and a second edge that are disposed opposite to each other, the heat dissipation region is adjacent to the first edge, and at least one of the one or more heat pipes extends from the heat dissipation region to a direction of the second edge.
8. The mobile terminal according to claim 7, wherein at least one of the one or more heat pipes is

not in contact with the battery.

9. The mobile terminal according to claim 8, wherein the middle frame comprises a top, a bottom disposed opposite to the top, and two sidewalls located between the top and the bottom, a gap is formed between the battery placement region and the side walls, and at least one of the one or more heat pipes extends into the gap.

10. The mobile terminal according to claim 7, wherein at least one of the one or more heat pipes is in contact with a surface of the battery.

11. The mobile terminal according to claim 7, wherein the one or more heat pipes comprise a first heat pipe and a second heat pipe that are disposed at intervals, the first heat pipe surrounds the battery placement region, and the second heat pipe overlaps the battery placement region.

12. The mobile terminal according to claim 7, wherein the middle frame comprises a top, a bottom disposed opposite to the top, and two sidewalls located between the top and the bottom, the heat dissipation region is located between the first edge of the battery placement region and the top, another element placement region is disposed between the second edge of the battery placement region and the bottom, the middle frame further comprises a second vapor chamber, the second vapor chamber is fastened in the another element placement region, and at least one of the one or more heat pipes is further connected to the second vapor chamber.

13. The mobile terminal according to claim 12, wherein the middle frame is provided with a first removed region, a second removed region, and a third removed region; the first vapor chamber is embedded in the first removed region, the second vapor chamber is embedded in the second removed region, and at least one of the one or more heat pipes is embedded in the third removed region; and the first removed region is a first groove or a hole, the second removed region is a second groove or a hole, and the third removed region is a third groove or a hole.

14. The mobile terminal according to claim 13, wherein one end of at least one of the one or more heat pipes overlaps the first vapor chamber, and the other end of the at least one of the one or more heat pipes overlaps the second vapor chamber.

15. The mobile terminal according to claim 1, wherein the middle frame is provided with a groove, the middle frame comprises a first overlapping side extending toward the inside of the groove, the first vapor chamber comprises another overlapping side, and the first overlapping side is fastened to the another overlapping side, so that the first vapor chamber is fastened to the middle frame.

16. The mobile terminal according to claim 15, wherein the first overlapping side and the another overlapping side are fastened through welding.

17. The mobile terminal according to claim 1, wherein the middle frame is provided with a groove, the first vapor chamber is accommodated in the groove, and the first vapor chamber is fastened to a bottom of the groove by using a heat conducting adhesive.

18. The mobile terminal according to claim 17, wherein the middle frame comprises a first overlapping side extending toward the inside of the groove, the first vapor chamber comprises a another overlapping side, and the first overlapping side is fastened to the another overlapping side.

19. The mobile terminal according to claim 1, wherein the middle frame is provided with a first removed region and another removed region, the another removed region is formed at the bottom of the first removed region, the first vapor chamber is embedded in the first removed region, at least one of the one or more heat pipes is partially embedded in the another removed region, a part of the at least one of the one or more heat pipes in the another removed region is stacked with the first vapor chamber, the first removed region is a groove or a hole, and the another removed region is a groove or a hole.

20. The mobile terminal according to claim 1, wherein: at least one of the one or more heat pipes extends between at least two of the electronic components, and/or at least one of the one or more heat pipes extends between at least one of the one or more electronic components and a boundary of the middle frame.

21. A middle frame assembly, configured to bear an electronic components in a mobile terminal,

wherein the electronic components comprises a heat source, the middle frame assembly comprising: a middle frame, one or more heat pipes, and a vapor chamber; the middle frame comprises a heat dissipation region corresponding to the heat source, and the vapor chamber is accommodated in the heat dissipation region and is configured to dissipate heat from the heat source; the one or more heat pipes are connected to the vapor chamber and are configured to dissipate heat from the vapor chamber; the mobile terminal includes the electronic components and an internal conducting wire; and a wire groove is disposed on an outer surface of at least one of the one or more heat pipes, the wire groove is configured to fasten the internal conducting wire, and the conducting wire is configured to electrically connect the electronic components.

22. The middle frame assembly according to claim 21, wherein at least one of the one or more heat pipes is further connected to a part that is in the middle frame and whose temperature is lower than a temperature of the vapor chamber.

23. The middle frame assembly according to claim 22, wherein at least one of the one or more heat pipes is connected to the middle frame by using a heat conducting medium.

24. The middle frame assembly according to claim 21, wherein the one or more heat pipes are connected to the vapor chamber by using a heat conducting medium.

25. A mobile terminal comprising a mainboard, a heat source and a middle frame assembly, wherein: the mainboard is installed on the middle frame assembly, and the heat source is disposed on the mainboard; the middle frame assembly comprises a middle frame, one or more heat pipes, and a first vapor chamber; the middle frame comprises a heat dissipation region corresponding to the heat source, and the first vapor chamber is accommodated in the heat dissipation region and is configured to dissipate heat from the heat source; the one or more heat pipes are connected to the first vapor chamber and are configured to dissipate heat from the first vapor chamber; wherein the mobile terminal further comprises a battery, the middle frame further comprises a battery placement region corresponding to the battery, the battery placement region comprises a first edge and a second edge that are disposed opposite to each other, the heat dissipation region is adjacent to the first edge, and at least one of the one or more heat pipes extends from the heat dissipation region to a direction of the second edge; the middle frame comprises a top, a bottom disposed opposite to the top, and two sidewalls located between the top and the bottom, the heat dissipation region is located between the first edge of the battery placement region and the top, another element placement region is disposed between the second edge of the battery placement region and the bottom, the middle frame further comprises a second vapor chamber, the second vapor chamber is fastened in the another element placement region, and at least one of the one or more heat pipes is further connected to the second vapor chamber; and the middle frame is provided with a first removed region, a second removed region, and a third removed region; the first vapor chamber is embedded in the first removed region, the second vapor chamber is embedded in the second removed region, and at least one of the one or more heat pipes is embedded in the third removed region; and the first removed region is a first groove or a hole, the second removed region is a second groove or a hole, and the third removed region is a third groove or a hole.
